

1. Open Source for Regulated Manufacturing

Welcome everyone. I'm Lukas Ott, Enterprise Architect, and today I want to show how open source can support scalable, inspectable, and GMP-ready IIoT data platforms.

The key message is simple: compliance does not come from a database alone.

It comes from the validated system around it: controlled ingestion, traceability, access control, retention, monitoring, testing, and lifecycle governance.

2. The Pressure: Connected Manufacturing Meets Regulation

Manufacturing is becoming more connected, more automated, and increasingly data-driven.

Machines, sensors, PLCs, vision systems, and edge devices generate high-frequency data. That data is no longer only useful for dashboards. In regulated manufacturing, machine data can become part of quality decisions, investigations, and inspection evidence.

That creates a tension: we want real-time data and scalable analytics, but we also need trustworthy records, controlled systems, and long-term inspection readiness.

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4. The Shopfloor Is Becoming a Data Platform

The shopfloor is no longer a collection of isolated machines. It is becoming a distributed data platform.

But many shopfloor architectures were built machine by machine, site by site, and vendor by vendor. That often leads to fragmented semantics, inconsistent data models, proprietary interfaces, and technology stacks that are difficult to validate end to end.

So the question for today is: how can we build a scalable IIoT data foundation without losing control, traceability, and trust?

5. The Core Tension

This is the core tension of the talk.

On one side, manufacturing wants more connectivity, more automation, more analytics, and eventually AI-supported decision-making.

On the other side, GMP environments require evidence. The system must show where data came from, when it was captured, whether it was changed, who or what changed it, and whether it can be retrieved when needed.

6. Regulatory Pressure Is Expanding

The pressure is not only more data. It is more connected systems, more cloud, more cybersecurity expectations, and more AI-supported processes.

For regulated manufacturing, the question is no longer only whether a single record is correct. The question is whether the full computerized system lifecycle is controlled.

That includes ingestion, storage, processing, access, monitoring, retention, and change management.

7. Major Aspects